

Empowering Education Through Ai-supported E-learning

R26-SE-041

Our Team



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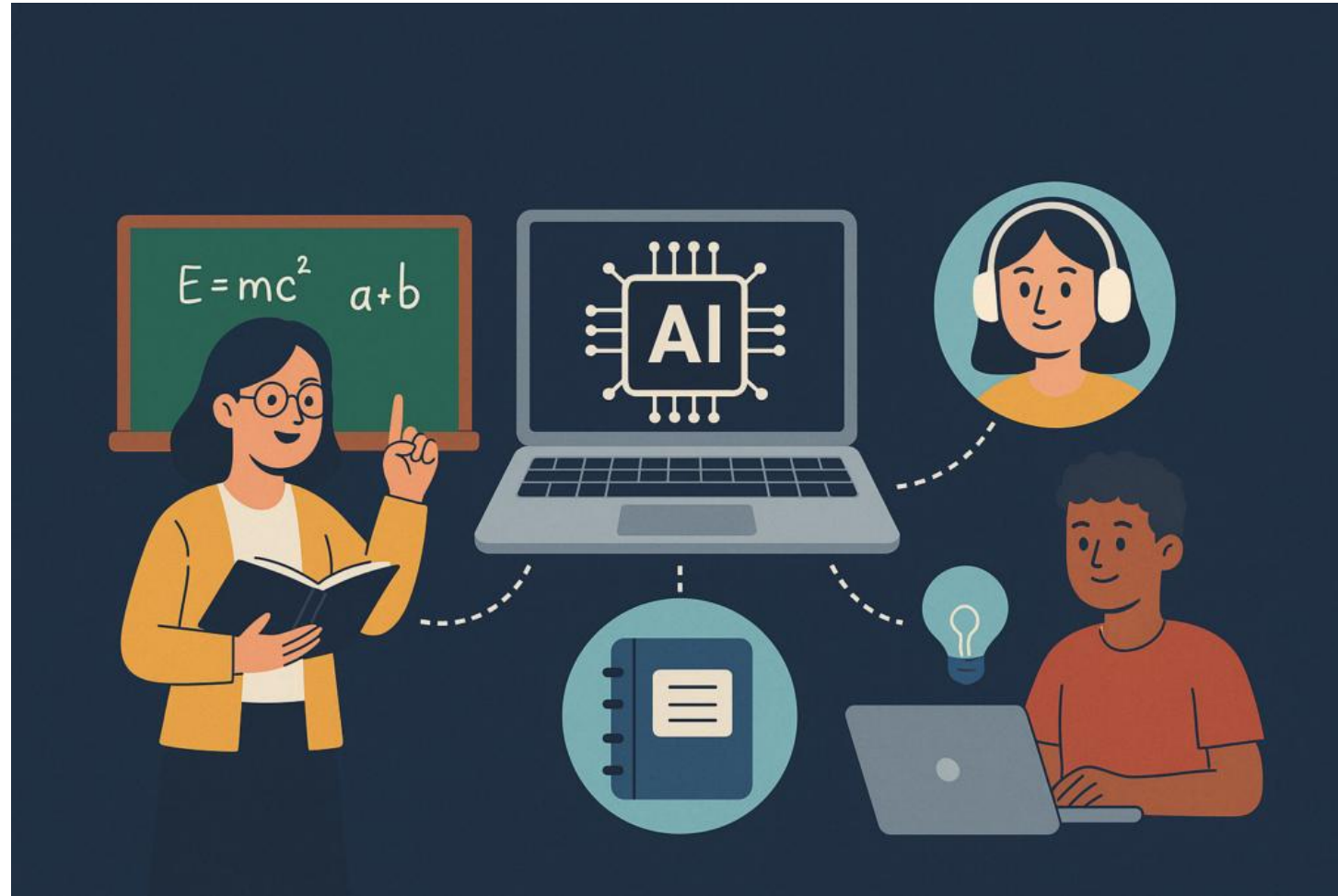


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Empowering secondary education through AI-Supported E-Learning



Empowering education through AI-Supported E-Learning

The Problem

- The standard "one-size-fits-all" teaching style doesn't work for everyone.
- It causes students to lose interest and miss important concepts.

Users & Impact

- Who it affects:** University students, teachers, and the universities.
- Social Impact:** Students fall behind and struggle to learn on their own.
- Economic Impact:** Students aren't ready for jobs on time, which hurts university success rates.

Current Tools & Their Failures

- Existing Tools:** Google NotebookLM, ChatGPT, Khanmigo.
- The Issues:** They only do basic tasks and lack deep personalization.
- The Gap:** They do not understand local languages like Sinhala and Tamil.

Our Proposed Solution

- What it is:** An all-in-one, AI-powered smart learning platform.
- How it works:** It brings together 4 unique AI tools to track student progress and automatically adjust the learning content and speed.



IT22264220 | KOJITHAN P.Y

Interactive Infographic and Visual Narrative Generator.

Turns student text notes into interactive, visual data stories. Helps students avoid information overload by making complex topics easy to see and explore.

Novelty (What makes it new?):

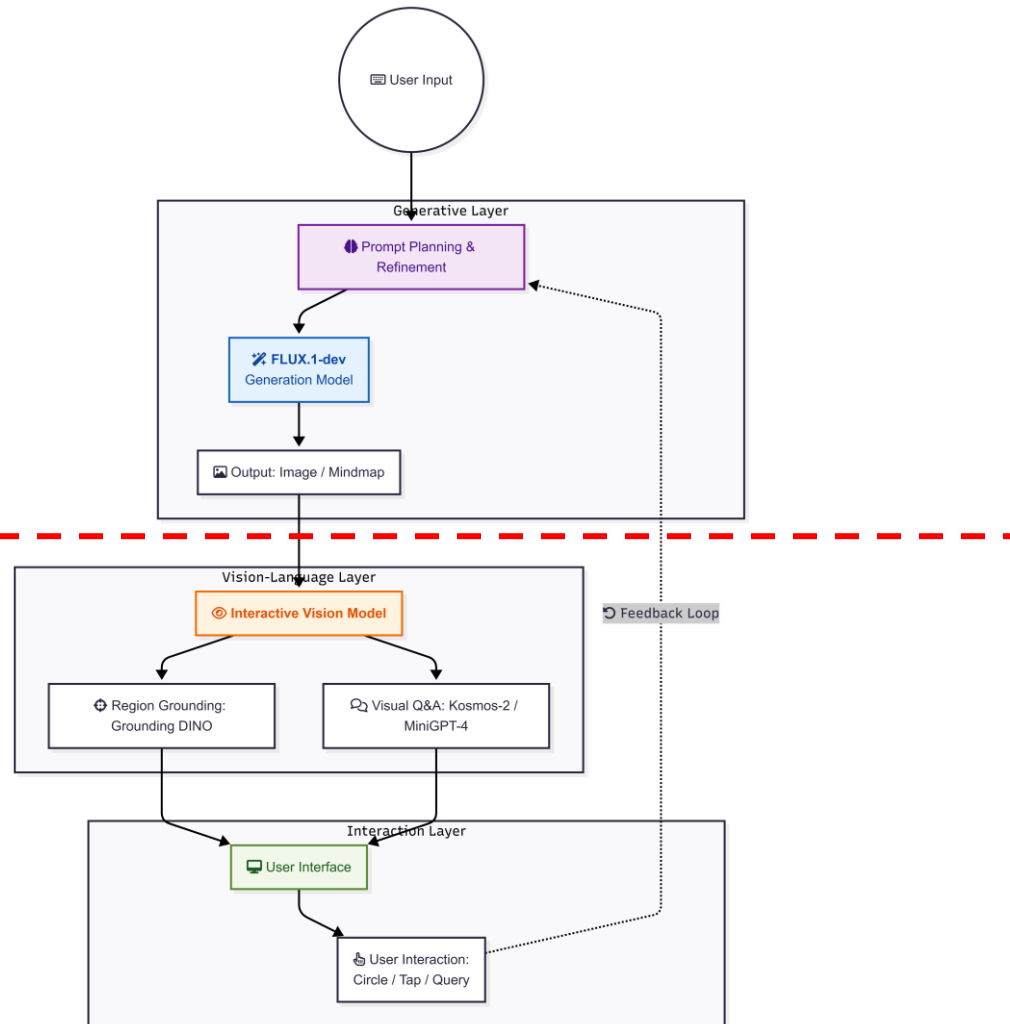
Existing tools only make flat, unclickable, non-interactive images.

My tool lets users click, explore, and find the exact source of the information.

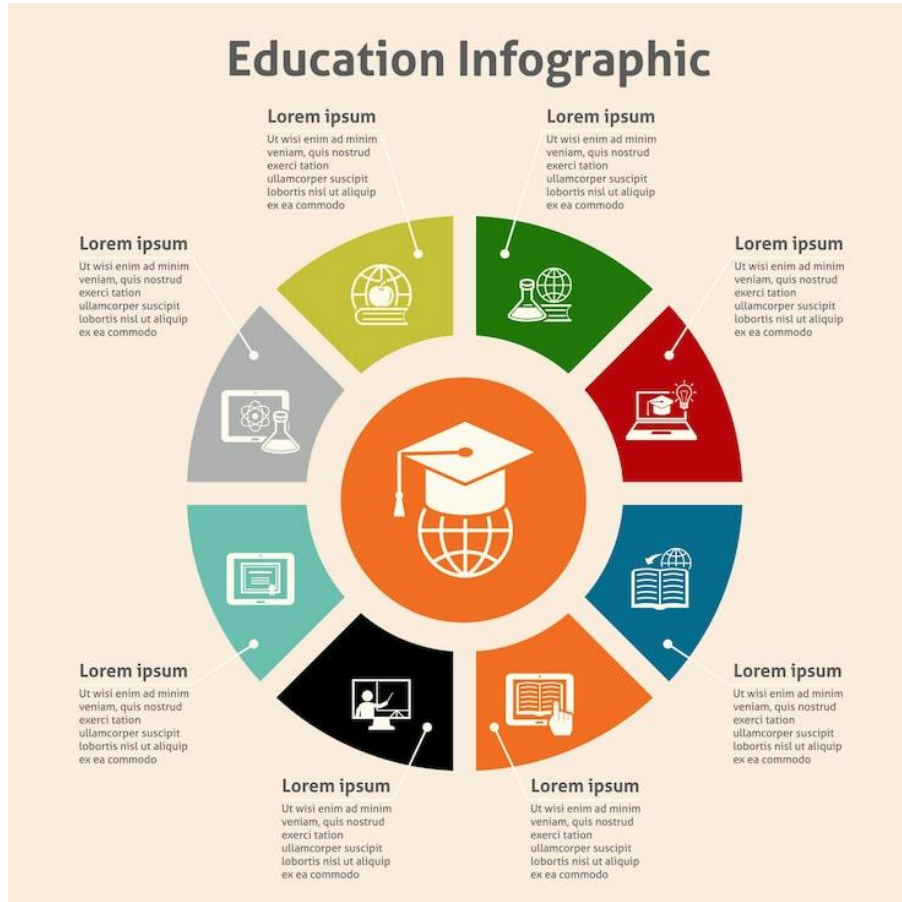
INTERACTIVE INFOGRAPHIC GENERATION FOR DL MODULE

Current Progress

1. Gather requirement and Find a dataset
2. Implemented a function to preprocess the data
3. Implemented a infograph generation model using a neural network
4. Evaluate the finetuned model using CLIP & FID score
5. Output the results



INTERACTIVE INFOGRAPHIC GENERATION FOR DL MODULE



An infographic is a visual representation of information, data, or knowledge designed to present complex information quickly and clearly, often using charts, graphs, and icons.

INTERACTIVE INFOGRAPHIC GENERATION FOR DL MODULE

Datasets

1. Synthetic Datasets (matplotlib.pyplot, ai, matplotlib.patches)
3. Infographics Dataset (Kaggle)
4. Francesco/activity-diagrams-qdobr
Training-259 V-45, Test-74 (HF)
5. shreyanshu09/Block_Diagram
Train 76.3k, Validation 8.66k
6. JasmineQiuqiu/diagrams_with_captions
Train 3.52k

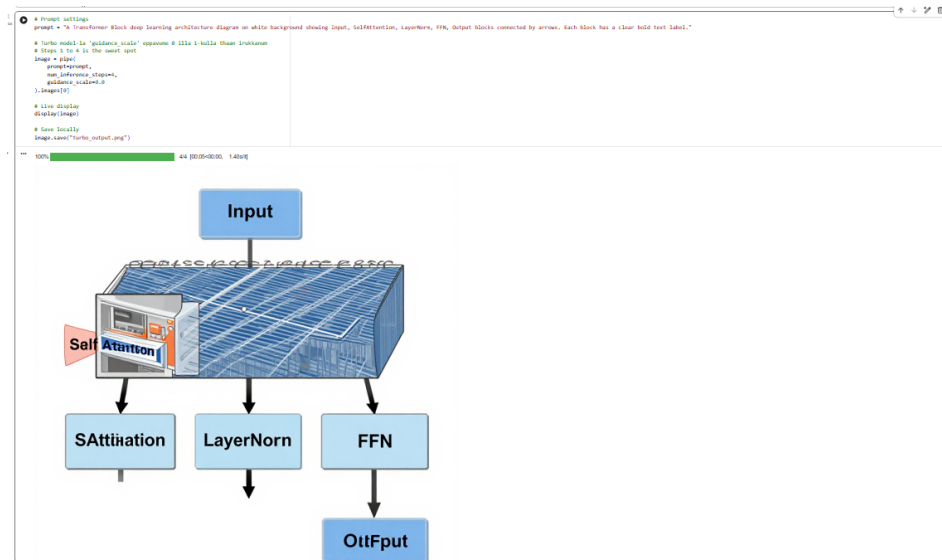
Preprocessing

1. Remove irrelevant images/ Datasets
2. Remove Black and white images
3. Remove low quality images
4. Convert into PNG
5. Convert into 1:1 ratio
6. Convert into 512x512
7. Labeling (Supervision, Autodistill)
8. Divide the dataset into 3 parts
(Training, Validation, Testing)

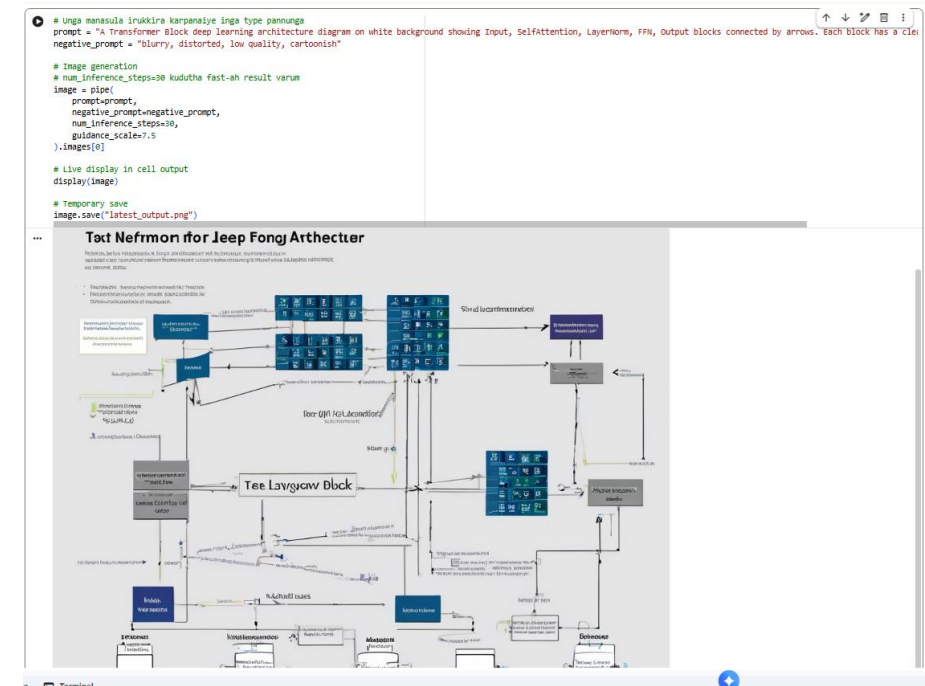
INTERACTIVE INFOGRAPHIC GENERATION FOR DL MODULE

Failed Models

Tongyi-MAI/Z-Image-Turbo – 6Billion



xinsir/controlnet-union-sdxl-1.0 - 1.25Billion

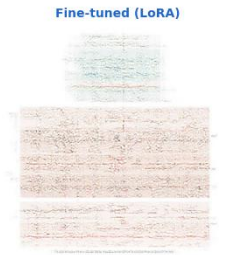


INTERACTIVE INFOGRAPHIC GENERATION FOR DL MODULE

Phase 1

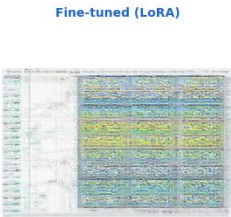
Input Prompt

A convolutional neural network architecture with input layer, three conv layers with ReLU activation, max pooling, and fully connected output layer



Input Prompt

ResNet-50 deep residual learning architecture diagram with skip connections and bottleneck blocks



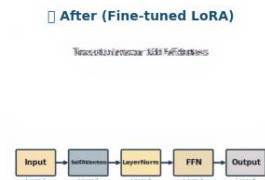
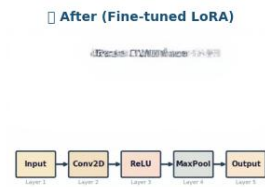
Input Prompt

Transformer architecture diagram showing encoder decoder structure with multi-head attention and feed forward layers



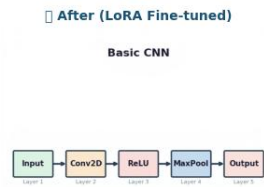
black-forest-labs/FLUX.1-schnell 12B

Phase 2



black-forest-labs/FLUX.1-dev 12B

Phase 3



black-forest-labs/FLUX.1-dev 12B

INTERACTIVE INFOGRAPHIC GENERATION FOR DL MODULE

Before and After Comparison

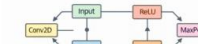
Input Prompt

A Basic CNN deep learning architecture diagram on white background showing Input, Conv2D, ReLU, MaxPool, Output blocks connected by arrows. Each block has a clear bold text label.

Before (Base Model)



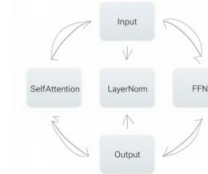
After (LoRA Fine-tuned)



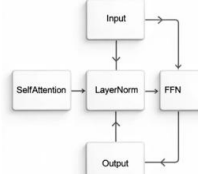
Input Prompt

A Transformer Block deep learning architecture diagram on white background showing Input, SelfAttention, LayerNorm, FFN, Output blocks connected by arrows. Each block has a clear bold text label.

Before (Base Model)



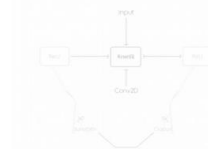
After (LoRA Fine-tuned)



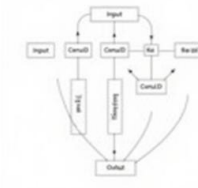
Input Prompt

A ResNet Block deep learning architecture diagram on white background showing Input, Conv2D, ReLU, Conv2D, Reshape, Conv2D, Output blocks connected by arrows with skip connection. Each block has a clear bold text label.

Before (Base Model)



After (LoRA Fine-tuned)



black-forest-labs/FLUX.1-dev 12B

INTERACTIVE INFOGRAPHIC GENERATION FOR DL MODULE

Model Accuracy and Training Analysis

[1/10] Base CLIP: 0.2431 | FT CLIP: 0.2923
100% ██████████ 20/20 [00:02<00:00, 6.97it/s]
100% ██████████ 20/20 [00:02<00:00, 6.96it/s]
[2/10] Base CLIP: 0.3030 | FT CLIP: 0.3515
100% ██████████ 20/20 [00:02<00:00, 6.95it/s]
100% ██████████ 20/20 [00:02<00:00, 6.94it/s]
[3/10] Base CLIP: 0.2532 | FT CLIP: 0.2088
100% ██████████ 20/20 [00:02<00:00, 6.94it/s]
100% ██████████ 20/20 [00:02<00:00, 6.92it/s]
[4/10] Base CLIP: 0.2496 | FT CLIP: 0.2716
100% ██████████ 20/20 [00:02<00:00, 6.95it/s]
100% ██████████ 20/20 [00:02<00:00, 6.94it/s]
[5/10] Base CLIP: 0.3437 | FT CLIP: 0.3251
100% ██████████ 20/20 [00:02<00:00, 6.93it/s]
100% ██████████ 20/20 [00:02<00:00, 6.91it/s]
[6/10] Base CLIP: 0.2709 | FT CLIP: 0.2332
100% ██████████ 20/20 [00:02<00:00, 6.93it/s]
100% ██████████ 20/20 [00:02<00:00, 6.93it/s]

FINAL EVALUATION SUMMARY

CLIP Score (Higher = Better)

Base Model : 0.2771

Fine-tuned Model: 0.2770

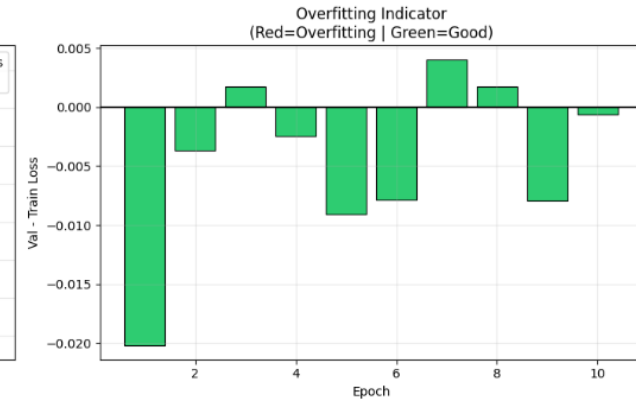
FID Score (Lower = Better)

Base Model : 222.4703

Fine-tuned Model: 214.9308

...

Training Analysis — FLUX.1-dev LoRA



==== TRAINING SUMMARY =====

Best Val Loss: 0.0138 at Epoch 9

Final Train: 0.0256

Final Val: 0.0250

Status: ✔ GOOD FIT

=====

INTERACTIVE INFOGRAPHIC GENERATION FOR DL MODULE

Technologies used

1. Python libraries
2. FastApi
3. Flux 1 Dev -12B
4. React
5. Gradio

Techniques used

1. Diffusion Models to flow-matching
2. 80:10:10 Rule for the Dataset
3. LoRA (r=16, 0.3289%, 16Bit)
4. Data Augmentation
5. 10 epochs (9th was the good fit)
6. CLIP Score & FID Score

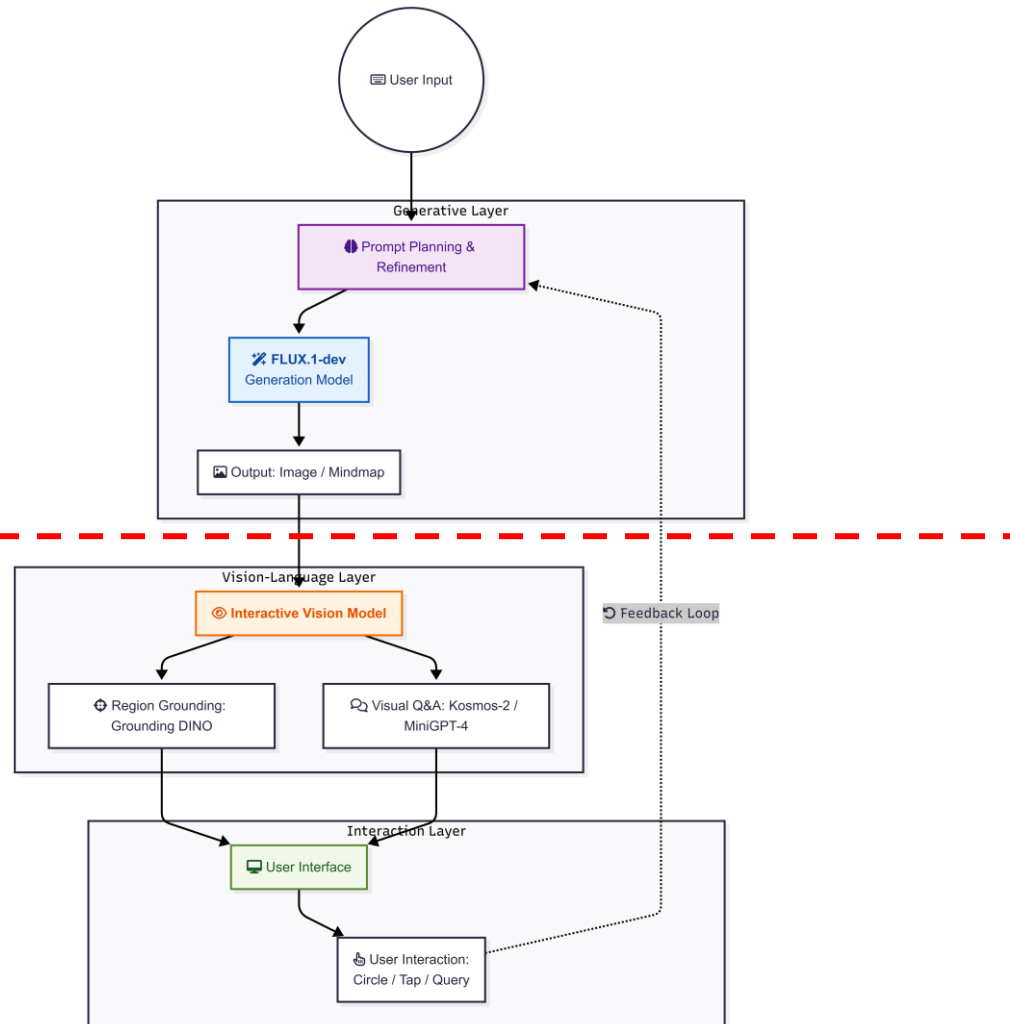
Tools and Resources used

1. Hugging Face
2. Kaggle
- 3. Google colab Pro - A100 GPU 80GB
(Spent 150 Units, LKR 6,607.54)**
3. Vast ai
4. modal.com

INTERACTIVE INFOGRAPHIC GENERATION FOR DL MODULE

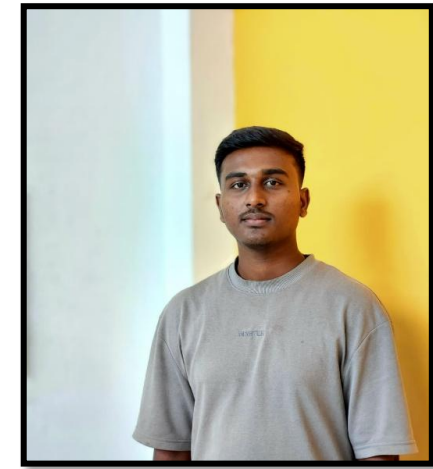
Next Progress

1. Improve the current model accuracy (CLIP & FID Score)
2. Implement a next phase (interactive infographic)
3. Implement the whole backend and frontend



INTERACTIVE AUDIO BASED DEEP LEARNING TUTOR

IT22172600 | BASKARAN V

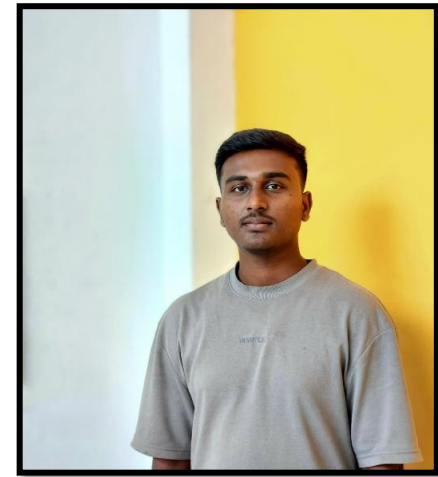


INTERACTIVE AUDIO BASED DEEP LEARNING TUTOR

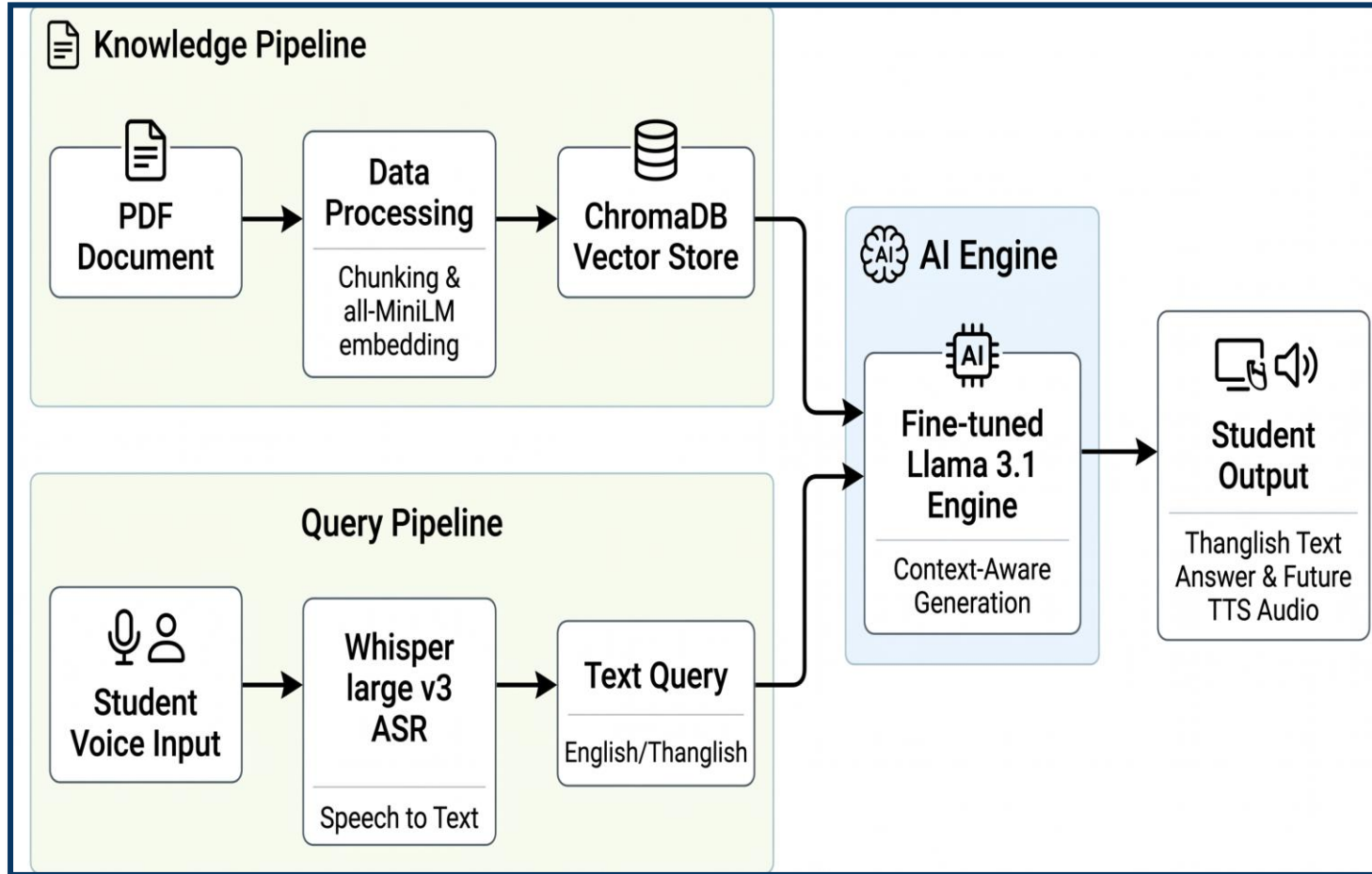
Contribution Of Your Component To The Overall Solution

1. Changes passive listening into an active learning experience
2. Allows real-time voice Q&A directly with the audio
3. Supports local languages like Sinhala and Tamil

Novelty: Unlike existing tools, this system allows real-time voice interactions using local languages like Sinhala and Tamil



INTERACTIVE DEEP LEARNING AUDIO BASED TUTOR



current process

- Gathered requirements and identified suitable datasets
- Implemented data preprocessing functions for cleaning and preparing the dataset
- Implemented the **Whisper Large V3 model** for speech-to-text processing
- Fine-tuned the **Llama 3.1 8B model** using custom datasets
- Generated and displayed the output results in text format

INTERACTIVE AUDIO BASED DEEP LEARNING TUTOR

Datasets

1. Sugumar/Task_2Translate_english_to_thanglish (Hugging face)
2. Thanglish Dataset (Kaggle)
3. gowtham58/Tamil_Thanglish_English
4. gowtham58/Tamil_Thanglish_English_lexicon
5. Synthetic Dataset (SLIIT Lecture Slides & Past Papers, Peradeniya Lecture Slides)

Preprocessing

1. Remove irrelevant contents in lecture
2. Data cleaning
3. Labeling
4. Divide the dataset into 3 parts (Training, Validation, Testing)

INTERACTIVE AUDIO BASED DEEP LEARNING TUTOR

CELL 4 — Load SLM (flan-t5-base) for Answer Generation

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

slm_name = "google/flan-t5-base"

tokenizer = AutoTokenizer.from_pretrained(slm_name)
slm_model = AutoModelForSeq2SeqLM.from_pretrained(slm_name)

print("✅ SLM (flan-t5-base) loaded!")
print("This is the RAG answer generation model.")
print("Fine-tuning will improve quality further (80% milestone).")

config.json: 1.40k? [00:00<00:00, 102kB/s]
tokenizer_config.json: 2.54k? [00:00<00:00, 184kB/s]
spiece.model: 100% 792k/792k [00:00<00:00, 3.97MB/s]
tokenizer.json: 2.42M? [00:00<00:00, 67.9MB/s]
special_tokens_map.json: 2.20k? [00:00<00:00, 177kB/s]
model.safetensors: 100% 990M/990M [00:07<00:00, 203MB/s]
Loading weights: 100% 282/282 [00:00<00:00, 716.46MB/s, Materializing param=shared.weight]
The tied weights mapping and config for this model specifies to tie shared.weight to lm_head.weight, but both are present in the checkpoints, so we will NOT tie them. You should update the config with 'tie_word_embeddings=False' in the config file.
generation_config.json: 100% 147/147 [00:00<00:00, 7.52kB/s]

✅ SLM (flan-t5-base) loaded!
This is the RAG answer generation model.
Fine-tuning will improve quality further (80% milestone).
```

Flan-t5-base- 250M

CELL 11 — Test Q&A: Ask in any language!

Try asking in Tamil, Sinhala, or English — system detects and responds accordingly.

```
# ---- Change this question to test different languages! ----
questions_to_test = [
    "What is a Convolutional Neural Network?", # English
    "CNN என்னால் என்ன?", # Tamil
    "What is max pooling in deep learning?", # English
]

for q in questions_to_test:
    print("=" * 60)
    res = ask_combined(q)
    print(f"Question : {res['question']}")
    print(f"Language : {res['detected_language']}")
    print(f"Answer : {res['answer']}")
    if res['pdf_sources']:
        print(f"PDF Ref : {res['pdf_sources'][0]}")
    if res['audio_sources']:
        print(f"Audio : {res['audio_sources'][0]['start']:.1f}s → {res['audio_sources'][0]['end']:.1f}s")
    print()

***
=====
Question : What is a Convolutional Neural Network?
Language : English
Answer : Convolutional Neural Network
PDF Ref : {'source': '2023-S2-SE4050-Lecture-02-Introduction to Artificial Neural Networks (V2).pdf', 'page': 28}
Audio : 0.0s → 4.3s

=====
Question : CNN என்னால் என்ன?
Language : Tamil
Answer :
PDF Ref : {'source': '2023-S2-SE4050-Lecture-02-Introduction to Artificial Neural Networks.pdf', 'page': 14}
Audio : 0.0s → 4.3s

=====
Question : What is max pooling in deep learning?
Language : English
Answer : Max pooling in deep learning refers to an algorithm that performs equally well on training data and any new samples or test dataset
PDF Ref : {'source': '2024-July-SE4050-Lecture-03-Basics on ANNs in Depth.pdf', 'page': 18}
Audio : 0.0s → 4.3s
```

CELL 2 — Load Whisper ASR Model

```
import whisper  
|  
whisper_model = whisper.load_model("small")  
  
print("✅ Whisper ASR model loaded!")  
print("    Model: small (supports Tamil, Sinhala, English)")
```

... 100% |██| 461M/461M [00:02<00:00, 167MiB/s]
✅ Whisper ASR model loaded!

Whisper Base model- 74M

Llama -3.1-Instruct -8B

pretrained and instruction tuned generative models in 8B, 70B and 405B sizes (text in/text out). The Llama 3.1 instruction tuned text only models (8B, 70B, 405B) are optimized for multilingual dialogue use cases and outperform many of the available open source and closed chat models on common industry benchmarks.

Model developer: Meta

Model Architecture: Llama 3.1 is an auto-regressive language model that uses an optimized transformer architecture. The tuned versions use supervised fine-tuning (SFT) and reinforcement learning with human feedback (RLHF) to align with human preferences for helpfulness and safety.

	Training Data	Params	Input modalities	Output modalities	Context length	GQA	Token count	Knowledge cutoff
Llama 3.1 (text only)	A new mix of publicly available online data.	8B	Multilingual Text	Multilingual Text and code	128k	Yes	15T+	December 2023
		70B	Multilingual Text	Multilingual Text and code	128k	Yes		
		405B	Multilingual Text	Multilingual Text and code	128k	Yes		

Supported languages: English, German, French, Italian, Portuguese, Hindi, Spanish, and Thai.

Llama 3.1 family of models. Token counts refer to pretraining data only. All model versions use Grouped-Query Attention (GQA) for improved inference scalability.

Model Release Date: July 23, 2024.

Status: This is a static model trained on an offline dataset. Future versions of the tuned models will

The screenshot shows the Hugging Face Inference Providers web interface. At the top, there's a navigation bar with the 'Inference Providers' logo and a 'NEW' badge. Below this, there's a 'Text Generation' tab and buttons for '+ New Chat' and 'Examples'. The main chat area has a dark background. A system message says 'Input a message to start chatting with meta-llama/Llama-3.1-8B-Instruct.' The user's prompt is 'Deep Learning-la 'Backpropagation' concept-ah oru chinna example-oda simple-ah explain pannu', displayed in a blue bubble. The model's response is 'Backpropagation (BP) la ninnu simple-ah explain pannalam. Backpropagation, enna? Enakku simple example pannalam.', displayed in a dark grey bubble. Below the response, there's a 'Backpropagation' section. At the bottom of the chat area, there's a text input field with the placeholder 'Your prompt here...' and a 'Send' button. Below the chat area, there are links for 'View Code Snippets' and 'JSON output', and a 'Compare providers' button. A section titled 'Model tree for meta-llama/Llama-3.1-8B-Instruct' shows a tree structure with 'Base model' as the root, which branches into 'Finetuned (1786)', 'Adapters', 'Finetunes', 'Merges', and 'Quantizations'. Each branch has a link to 'this model' and a count of models. For example, 'Finetuned' has 1786 models, 'Adapters' has 2143 models, 'Finetunes' has 2715 models, 'Merges' has 92 models, and 'Quantizations' has 627 models. At the bottom, there's a section titled 'Spaces using meta-llama/Llama-3.1-8B-Instruct' with a count of 100. The footer shows the user's name 'suraj291/Survival_Island_Game' and the provider 'artificialguybr/CantRunThisLLM'.

INTERACTIVE AUDIO BASED DEEP LEARNING TUTOR

The screenshot shows the Hugging Face website for the meta-llama/Llama-3.1-Instruct model. The page includes a header with navigation links, a main content area with a table of training data, supported languages, and model release date, and a right sidebar with an 'Inference Providers' section.

Model developer: Meta

Model Architecture: Llama 3.1 is an auto-regressive language model that uses an optimized transformer architecture. The tuned versions use supervised fine-tuning (SFT) and reinforcement learning with human feedback (RLHF) to align with human preferences for helpfulness and safety.

	Training Data	Params	Input modalities	Output modalities	Context length	GQA	Token count	Knowledge cutoff
Llama 3.1 (text only)	A new mix of publicly available online data.	8B	Multilingual Text	Multilingual Text and code	128k	Yes	15T+	December 2023
		70B	Multilingual Text	Multilingual Text and code	128k	Yes		
		405B	Multilingual Text	Multilingual Text and code	128k	Yes		

Supported languages: English, German, French, Italian, Portuguese, Hindi, Spanish, and Thai.

Llama 3.1 family of models. Token counts refer to pretraining data only. All model versions use Grouped-Query Attention (GQA) for improved inference scalability.

Model Release Date: July 23, 2024.

Status: This is a static model trained on an offline dataset. Future versions of the tuned models will

Inference Providers

Text Generation + New Chat Examples

Backpropagation

Backpropagation, enna? Enna? Enakku simple explanation pannalam.

- Neural Network:** Nee, namma neural network la, input, hidden layer, output layer ayya. Input layer la, input values ayya. Hidden layer la, input values ki idhalam. Output layer la, final output ayya.
- Error Calculation:** Nee, namma neural network la, output layer la, error ayya. Error, enna? Enna? Enakku simple explanation pannalam. Error, enna? Enna? Enakku simple explanation pannalam.

Your prompt here... Send

View Code Snippets JSON output Compare providers

Model tree for meta-llama/Llama-3.1-Instruct

- Base model meta-llama/Llama-3.1-8B
 - Finetuned (1786) this model
 - Adapters 2143 models
 - Finetunes 2715 models
 - Merges 92 models
 - Quantizations 627 models

Spaces using meta-llama/Llama-3.1-8B-Instruct 100

- sura291/Survival_Island_Game
- artificialguybr/CaniRunThisLLM

Research Project | Llama 3.1 8B | LoRA v2 | Whisper large-v3 | ChromaDB RAG

Upload Learning Material

Ask by Voice

Ask by Text

System Status

Step 2 (alternative) - Type your question directly

Useful as a backup during your viva demo. Same Hybrid RAG pipeline. Language auto-detected.

Your Question (English or Thanglish)

backpropagation epidy work pannum endu sollunga?

Generate Answer

Clear

AI Answer

Backpropagation la previous layers ku error share panna nadakkum. So each layer learn correct output produce panna nu understand pannum. Athanala weights update panni model improve aagum.

Retrieved Context / Source Evidence

Retrieved Sources:

* 2023-S2-SE4050-Lecture-02-Introduction to Artificial Neural Networks V2.pdf -- Page 23

Drive

Get answers from Drive

My Drive > Research_09 > datasets

Ask Gemini Summarise, analyse and get up to speed with files in this folder.

Type People Modified Source

Name	Owner	Date modified	File size	Sort
train_combined.jsonl	me	9 May me	263 KB	
test_combined.jsonl	me	9 May me	47 KB	

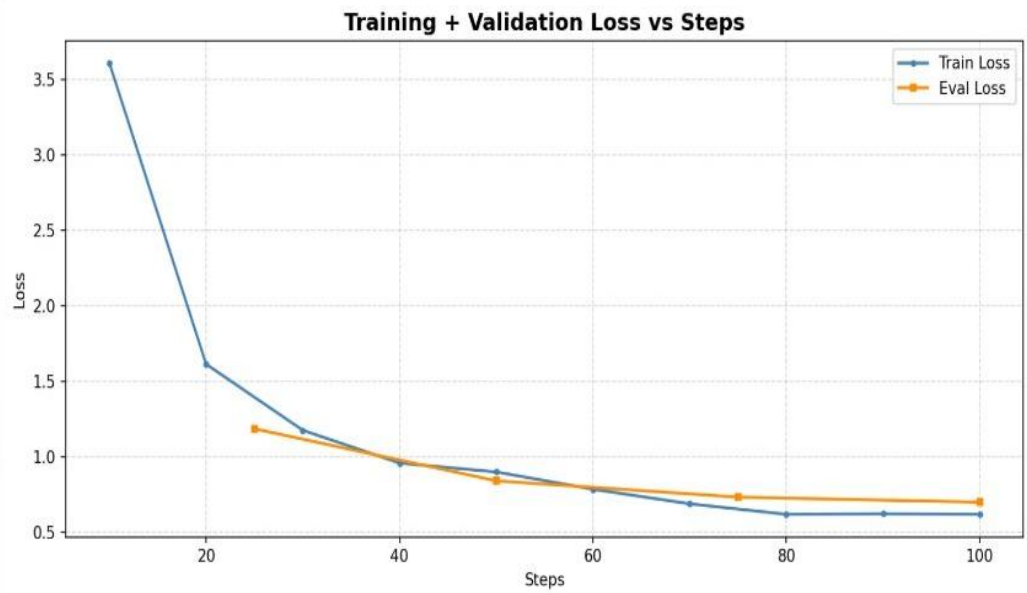


My Drive > Research_08 > datasets

Ask Gemini Summarise, analyse and get up to speed with files in this folder.

Type People Modified Source

Name	Owner	Date modified	File size	Sort
viva_style.jsonl	me	8 May me	68 KB	
variation_style.jsonl	me	8 May me	61 KB	
train.jsonl	me	8 May me	172 KB	
test.jsonl	me	8 May me	20 KB	
conversation_style.jsonl	me	8 May me	59 KB	
enn_master_dataset.jsonl	me	8 May me	192 KB	



INTERACTIVE AUDIO BASED DEEP LEARNING TUTOR

Technologies used

1. Python libraries
2. FastApi
3. meta-llama/Llama-3.1-8B-Instruct
4. React
5. Gradio
6. Diffusion Models
7. 80:10:10 Rule for the Dataset
8. LoRA
9. QLoRA (4 Bit Quantization)

Tools and Resources used

1. Hugging Face
2. Kaggle
- 3. Google colab Pro - A100 GPU 80GB
(Spent 150 Units, LKR 6,607.54)**
4. modal.com

IT22223876 | NISHARA T



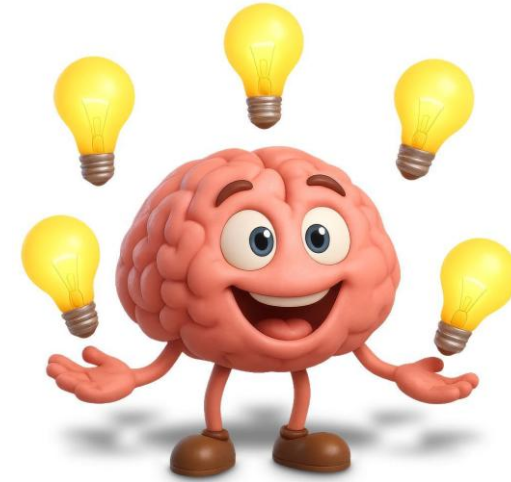
Deep Learning Powered Adaptive Quiz and Assessment System

Deep Learning Powered Adaptive Quiz and Assessment System

Solving the Sub-Problem:



- DL students at SLIIT struggle with MCQ exam preparation due to misalignment between lectures and past papers.
- Students don't have personalized, time-aware study plans and tools to prioritize high-impact MCQs from historical papers hinders effective revision and outcomes.



- This NLP system analyzes lecture notes to generate exam-standard MCQs and provides a guided learning experience through progressive hints. It bridges the gap between lectures and exams, ensuring students are conceptually ready for any complex scenario.

Deep Learning Powered Adaptive Quiz and Assessment System

Meeting User Requirements:

Cognitive Support: Guided learning through tiered hints (Hard/Medium/Easy) that mimic a human tutor.

Knowledge Tracing: Automatically adjusts the difficulty of the next question based on the student's attempt counts.

Positive Learning Environment: Students need resources (like videos or articles) to fix their mistakes immediately.



Deep Learning Powered Adaptive Quiz and Assessment System

Design Excellence / Contribution

Efficient Fine-Tuning (LoRA): Instead of training a massive AI from scratch, I used Parameter-Efficient Fine-Tuning to teach the Qwen2-7B model the specific Socratic hinting format.

Robust Parsing Architecture: I engineered a custom Regex-based extraction system to consistently and accurately pull multi-part data (Questions, Hints, Links) from the AI's generated text.

4-Bit Quantization: I applied BitsAndBytes technology to compress the heavy 7-Billion parameter model. This ensures the engine runs highly efficiently on standard hardware without losing its logical reasoning.

Deep Learning Powered Adaptive Quiz and Assessment System

User Feedback on Prototype:

- The progressive hints are excellent. The 'Hard' level forced me to logically think and re-read my notes rather than just guessing

- Users noted that the AI generation takes about 20 seconds per question, which is expected for heavy 7B models but slightly delays the quiz flow.
- The automated study recommendation links (YouTube/GeeksForGeeks) occasionally hallucinate and generate broken or inactive URLs.



Deep Learning Powered Adaptive Quiz and Assessment System

Customer Persona:

- **University Students:** Learners needing guided, logical exam preparation.
- **Academic Educators:** Teachers looking to automate smart quiz generation from notes.



Deep Learning Powered Adaptive Quiz and Assessment System

What is Unique in Your Solution?

- **Socratic Hinting Engine:** Standard quizzes just give the right answer. Our engine gives 3-tier progressive hints (Hard, Medium, Easy) to make students think and learn.
- **Adaptive Knowledge Tracing:** The system doesn't give the same test to everyone. It automatically changes the difficulty of the next question based on the student's attempt count.
- **Smart Resource Recommendations:** Instead of giving negative "remedial" marks, the system automatically shares exact YouTube and GeeksForGeeks links for the specific topics the student is weak in.

How will the cost be recovered?

Purchased 200 Google Colab compute units at a cost of Rs. 6,600 per member.

Deep Learning Powered Adaptive Quiz and Assessment System

Datasets



AI generated datasets: Synthetic JSON datasets



SLIIT Lecture Slides & Past Papers: Deep Learning module PDFs and historical past paper documents.

Peradeniya Lecture Slides: Acquired a limited subset (only 3 slide decks). Due to insufficient data volume, they provided minimal contribution to model training and were used solely for initial syllabus comparison.

Deep Learning Powered Adaptive Quiz and Assessment System

Preprocessing

- **Remove irrelevant slides:** Filtered and removed irrelevant lecture slides collected from the Peradeniya students' folder.
- **Extract Raw Data:** Extracted raw text content from the remaining SLIIT lecture PDFs and historical past papers.
- **Data Cleaning:** Removed noisy characters, headers, and footers from the raw text to generate **Cleaned Data**.
- **Socratic Formatting:** Mapped the cleaned data into the strict Socratic Prompt Template (Instruction → Context → Question + Hints).
- **Tokenization:** Converted the text into numerical IDs using the Qwen2-7B AutoTokenizer.

DEEP LEARNING POWERED ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

Current Progress (50%)

1. Find a datasets and gather requirement
2. Implemented a MCQ generation model
3. Implemented a function that extract quiz from the text
4. Implemented difficulty level hints
5. Output the results

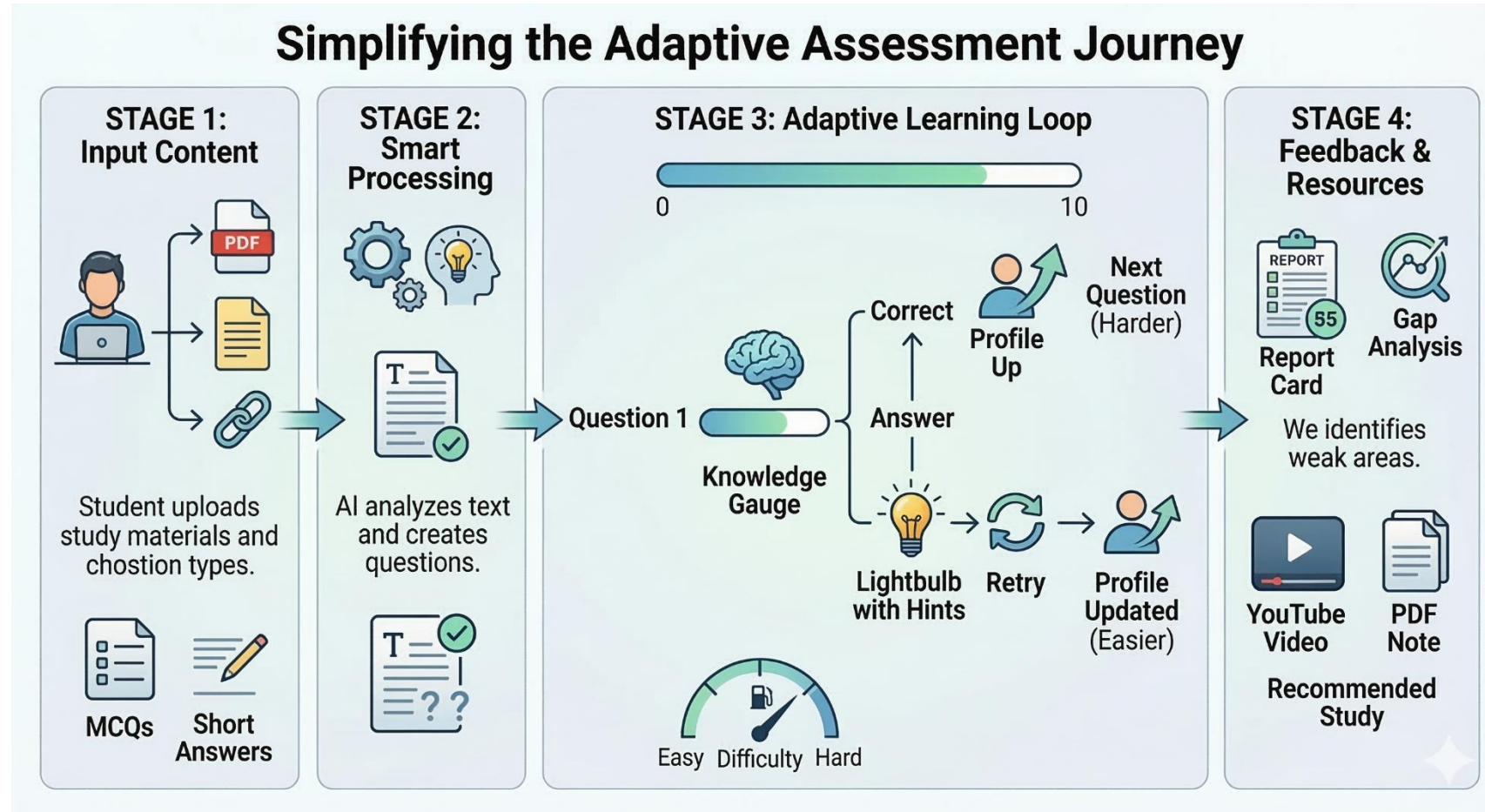
DEEP LEARNING POWERED ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

Next Expected Progress (100%)

1. Optimize model interface to reduce the generation latency below 20 seconds.
2. Refine the recommendation engine to resolve URL hallucinations in study links
3. Implement a PDF parsing module to allow users to directly upload lecture documents instead of manual text pasting.
4. Expand the engine's capabilities to generate Socratic "Essay" formats to test diverse cognitive levels.

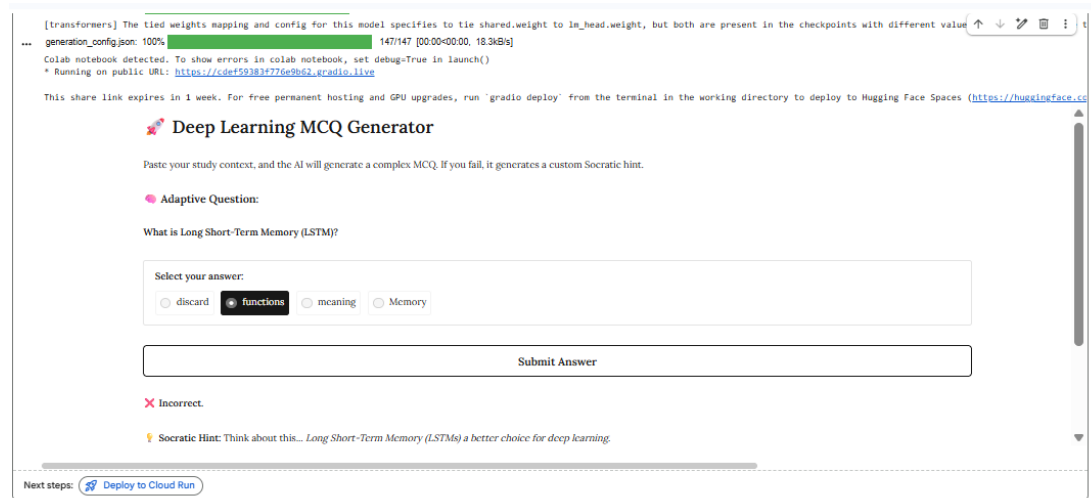
DEEP LEARNING POWERED ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

SYSTEM OVERVIEW



DEEP LEARNING POWERED ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

Model Rejection Base T5 model

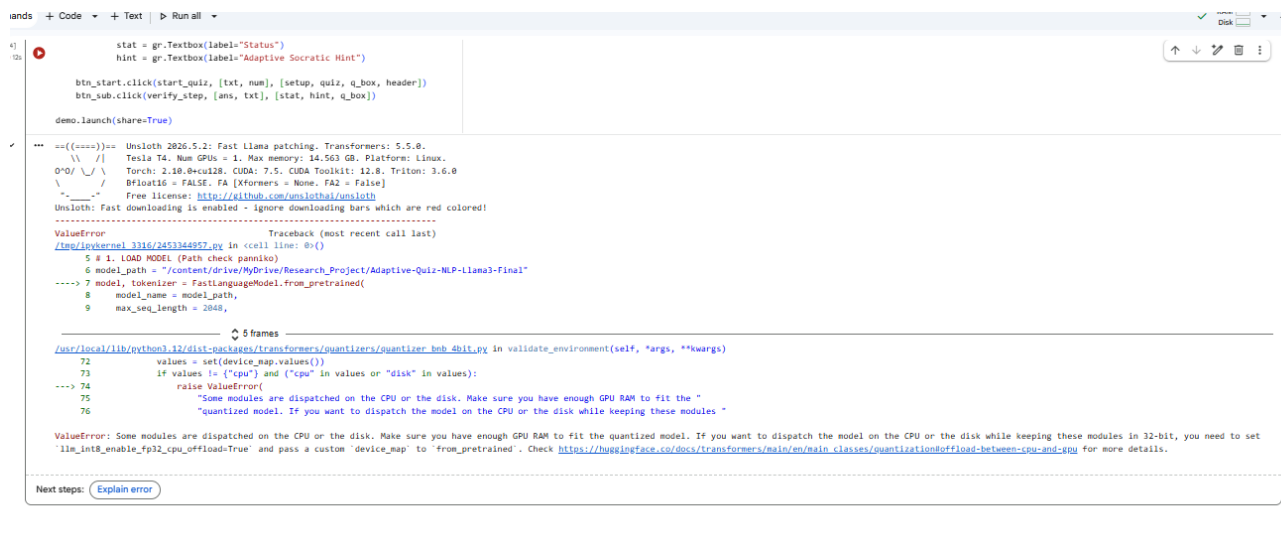


Base T5 was rejected due to its inability to generate contextually relevant distractors, producing random, easily guessable options (e.g., 'discard' for LSTM). It fails to meet the rigorous academic standards required for a University-level Adaptive Assessment.

DEEP LEARNING POWERED ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

Model Rejection

unsloth/llama-3-8b-bnb-4bit



```
stat = gr.Textbox(label="Status")
hint = gr.Textbox(label="Adaptive Socratic Hint")

btn_start.click(start_quiz, [txt, num], [setup, quiz, q_box, header])
btn_sub.click(verify_step, [ans, txt], [stat, hint, q_box])

demo.launch(share=True)

==((====))== Unsloth 2026.5.2: Fast Llama patching. Transformers: 5.5.0.
\\ /| Tesla T4. Num GPUs = 1. Max memory: 14.563 GB. Platform: Linux.
O/O/ \ / \ Torch: 2.10.0+cu128. CUDA: 7.5. CUDA Toolkit: 12.8. Triton: 3.6.0
\ ____ / Bfloat16 = FALSE. FA [Xformers = None, FA2 = False]
Free license: https://github.com/unslothai/unsloth
Unsloth: Fast downloading is enabled - ignore downloading bars which are red colored!
-----
ValueError                                Traceback (most recent call last)
/home/llmuser/.cache/unsloth/20260502 in cell line: 0()
5 # 1. LOAD MODEL (Path check panniko)
6 model_path = "/content/drive/MyDrive/Research_Project/Adaptive-Quiz-NLP-Llama3-Final"
----> 7 model, tokenizer = FastLanguageModel.from_pretrained(
8     model_name = model_path,
9     max_seq_length = 2048,

5 frames
/usr/local/lib/python3.12/dist-packages/transformers/quantizers/quantizer_bnb_4bit.py in validate_environment(self, *args, **kwargs)
72     values = set(device_map.values())
73     if values != {"cpu"} and {"disk"} in values:
----> 74         raise ValueError(
75             "Some modules are dispatched on the CPU or the disk. Make sure you have enough GPU RAM to fit the "
76             "quantized model. If you want to dispatch the model on the CPU or the disk while keeping these modules "
```

Llama-3 8B exhibited high variability in response structures (e.g., inconsistent labeling of MCQ options and hints). This unpredictability led to frequent automated parsing errors in the Gradio-based interface, resulting in blank UI elements and a degraded user experience.

DEEP LEARNING POWERED ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

Selected Model

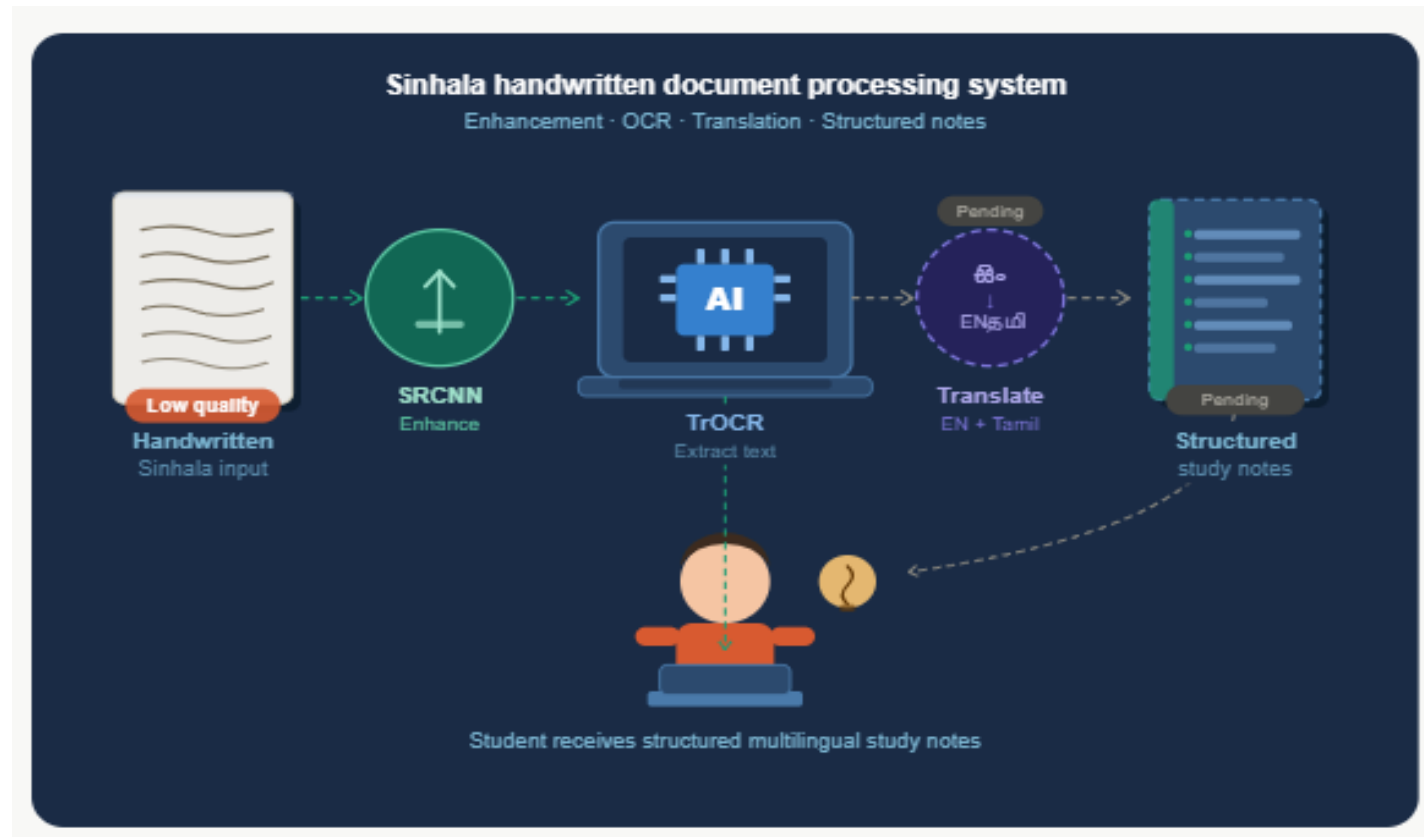
Qwen2-7B-Instruct

- **Advanced Reasoning:** Delivers superior logical deduction (better than Llama-3) required for generating 3-tier Socratic hints.
- **Strict Formatting:** Consistently maintains the complex JSON template (Question → Hints → Answer) without breaking code.
- **Massive Context Window:** Features a 128K token capacity, easily processing full lecture PDFs and past papers at once.
- **Hardware Efficiency:** Highly optimized for 4-bit quantization, allowing efficient fine-tuning on standard local GPUs.

Multiple Language support

SARMITHA. S – IT22637482

Sinhala Handwritten Image Enhancement Using Deep Learning for Improved Text Extraction and Study Material Generation



Sinhala Handwritten Image Enhancement Using Deep Learning for Improved Text Extraction and Study Material Generation

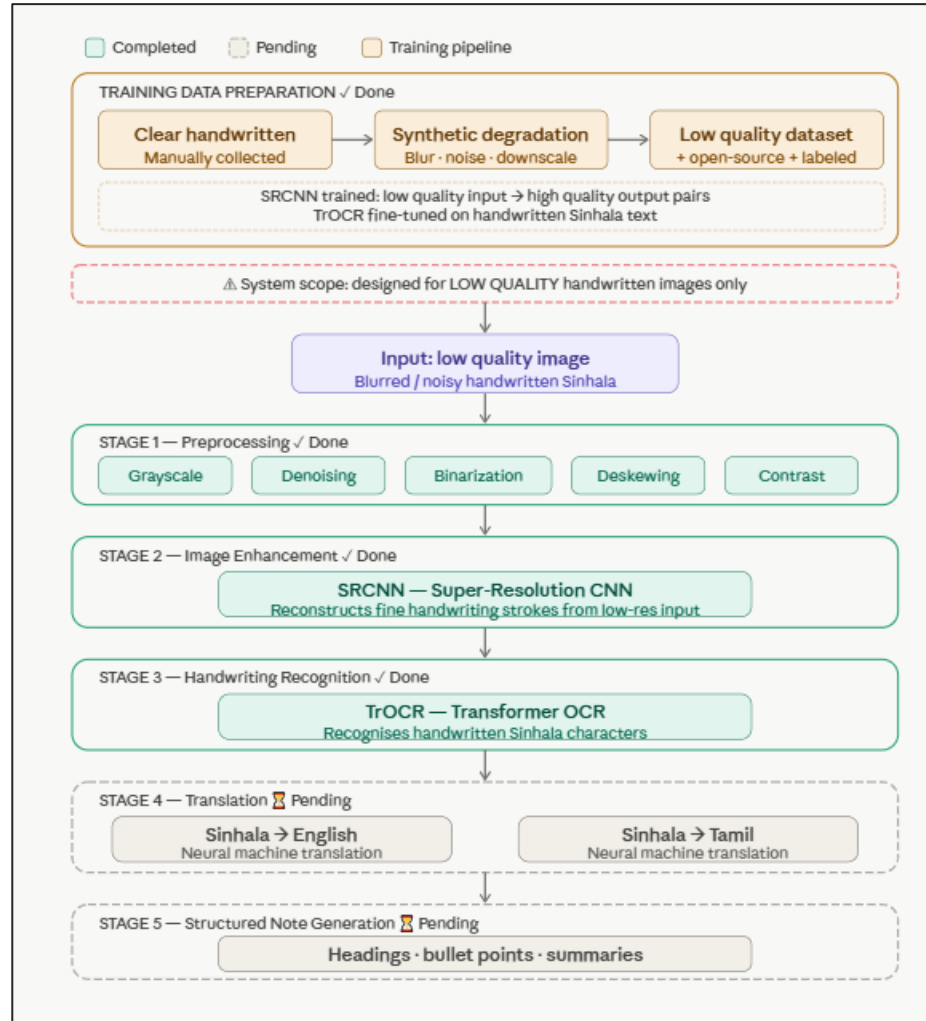
Research Gap

- Existing tools work well on printed/clean text but fail to handwritten images.
- Existing tools work well on English and some other languages but often fail for Sinhala language.
- Most existing tools work well in clean high quality images but fail to low quality images.
- No end-to-end system exists combining enhancement + OCR + translation + structured output for Sinhala handwriting.

Novelty

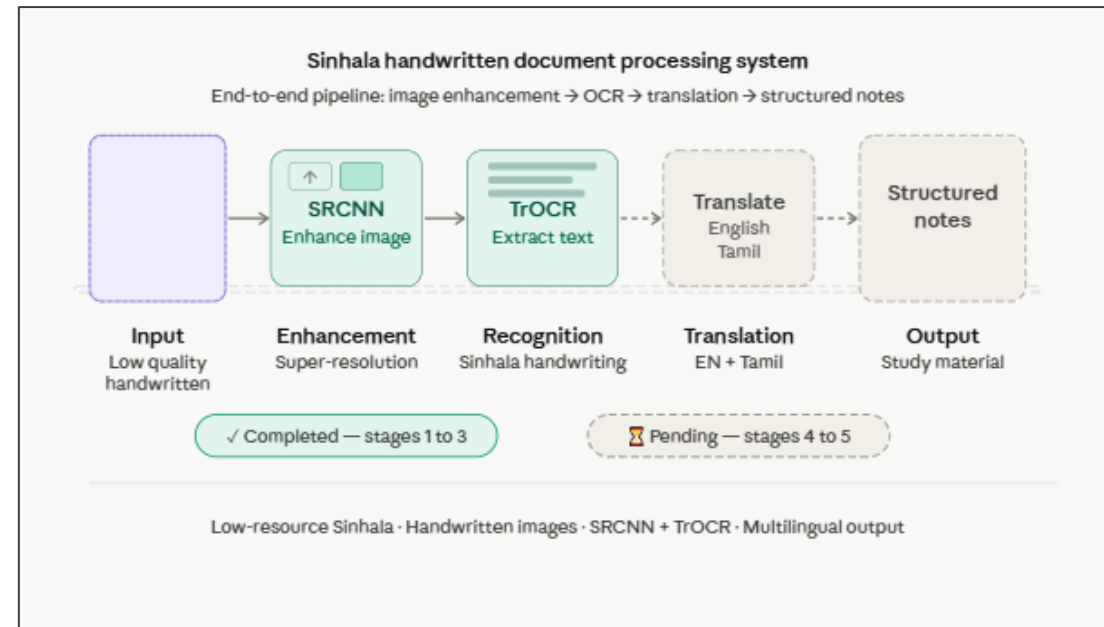
- My system works on low quality handwritten images.
- Adapts a text recognition approach specifically for low-resource Sinhala handwriting data
- Provides a complete pipeline that converts handwritten notes into structured, multilingual study material (English and Tamil)

System overview diagram

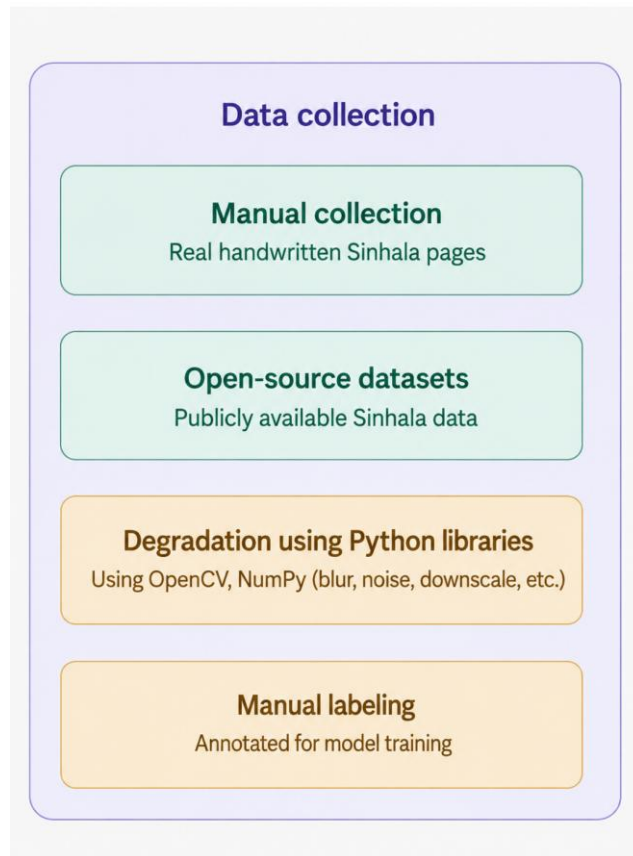


Completed

Pending



Datasets Collection Methods



Preprocessing Methods

Downsampling

Upsampling

Gaussian Blur

Noise Addition

Library used

cv2 (OpenCV)

cv2 (OpenCV)

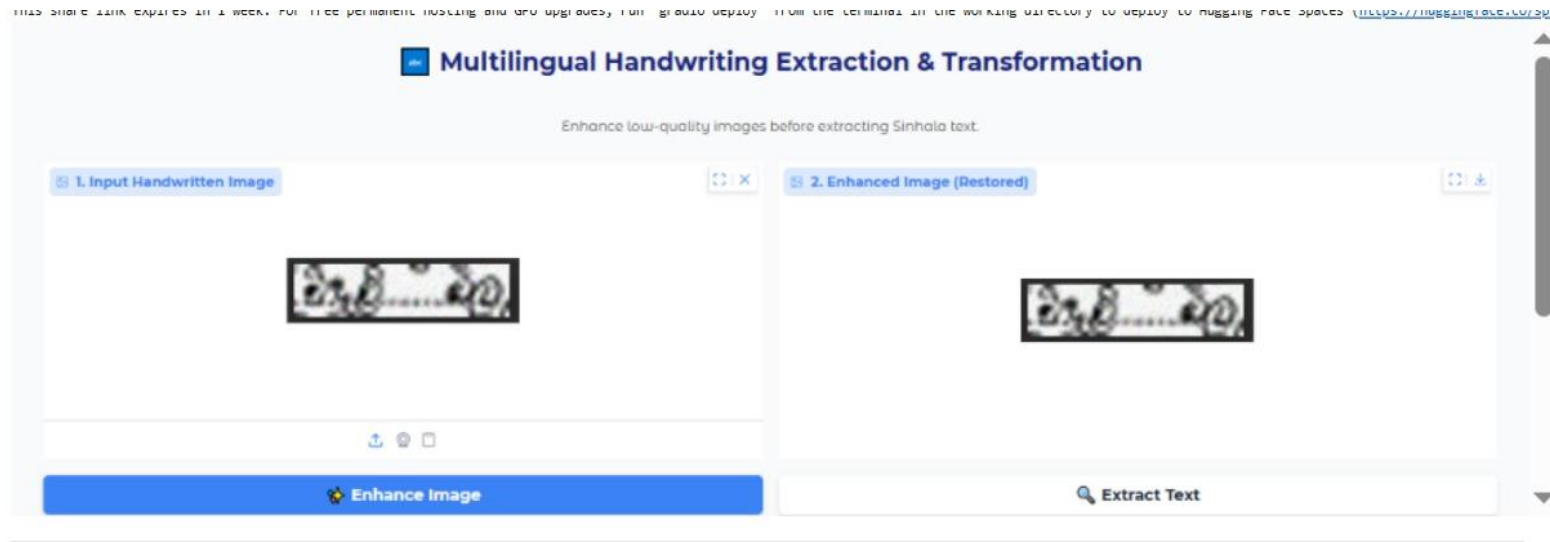
cv2 (OpenCV)

numpy

Sinhala Handwritten Image Enhancement Using Deep Learning for Improved Text Extraction and Study Material Generation

Failed Models

ESRGAN – Low Quality Enhancement

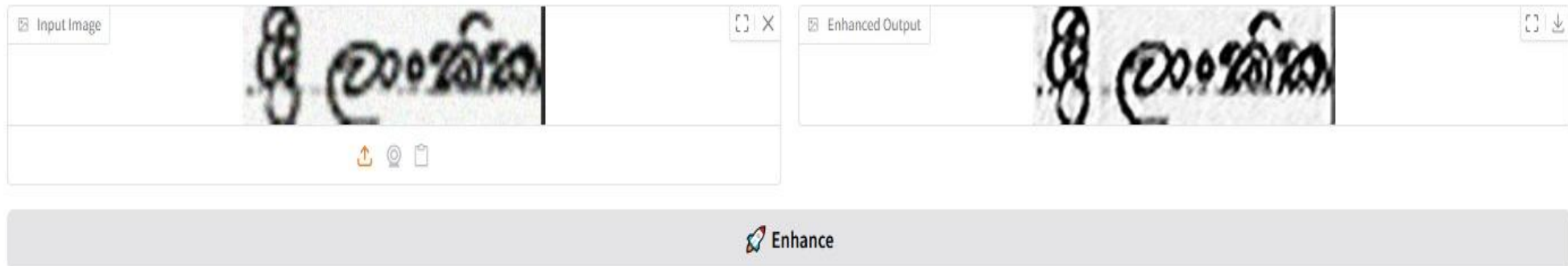


Sinhala Handwritten Image Enhancement Using Deep Learning for Improved Text Extraction and Study Material Generation

Selected Models

SRCNN – Low Quality Enhancement

🔥 SRCNN Enhancement (Original Size Output Fixed)



Sinhala Handwritten Image Enhancement Using Deep Learning for Improved Text Extraction and Study Material Generation

TrOCR - For Sinhala Extraction



Bought,

Google Colab – Rs.6600
200 units

Empowering education through AI-Supported E-Learning

- Target Market:** The EdTech sector, specifically focusing on Higher Education and Lifelong Learning.
- Customer Profile:** University undergraduates who struggle with cognitive overload , and educators requiring AI-assisted course authoring tools.

09. Commercialization Aspects

SaaS – B2B & B2C



BUSINESS MODEL

Individual



RS. 1000/month

- ✓ Feature Updates
- ✓ Ongoing Technical Support

Family



RS. 2500/month

- ✓ Up to 3 students
- ✓ Multi Student Management
- ✓ Feature Updates
- ✓ Ongoing Technical Support

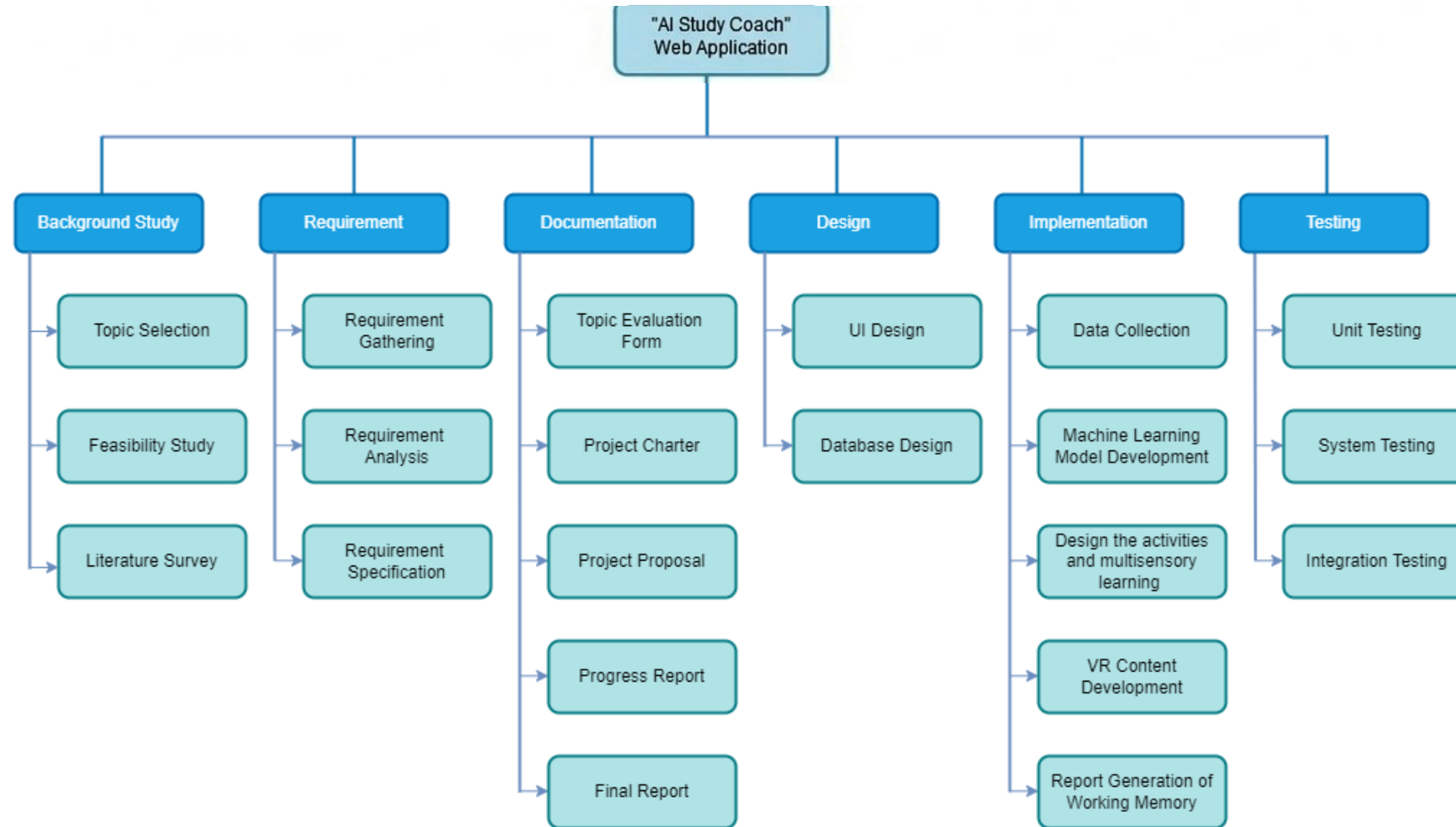
Classroom



RS. 5000/month

- ✓ Up to 40 students
- ✓ Multi Student Management
- ✓ Feature Updates
- ✓ Ongoing Technical Support

Work Breakdown Chart



Empowering education through AI-Supported E-Learning

PROJECT TIMELINE & GANTT CHART: RESEARCH PROPOSAL & REPORT (2026)

